Database Normalization: Practice Questions

First, Second, and Third Normal Forms (1NF, 2NF, 3NF)

FIRST NORMAL FORM (1NF)

A relation is in 1NF if all attributes contain only atomic (indivisible) values.

Question 1: Easy

The following table stores student course enrollment data. Identify the violation of 1NF and decompose the table to satisfy 1NF.

StudentID	StudentName	Courses
S001	Alice Johnson	Math, Physics, Chemistry
S002	Bob Smith	History, English
S003	Carol White	Biology, Chemistry, Math

What is the normalization violation?

Decompose the table to 1NF: (Create normalized table structure)

Question 2: Medium

Analyze the following employee project assignment table. Identify 1NF violations and normalize it. Note that some employees work on multiple projects with different hours.

EmployeeID	EmployeeName	Department	Projects & Hours
E001	John Doe	IT	ProjectA(40h), ProjectB(10h)
E002	Jane Wilson	HR	ProjectC(35h)
E003	Mike Brown	IT	ProjectA(20h), ProjectC(15h), ProjectD(5h)

Questions:

• What attributes violate 1NF?

• Design the normalized tables required.

Question 3: Slightly Difficult

The table below tracks library book reservations. Some books have multiple authors, cover multiple genres, and have multiple reservation dates. Normalize the table to 1NF and explain your decomposition strategy.

BookID	Title	Authors	Genres	ReservationDates
B001	Data Science Handbook	Smith, Johnson, Lee	Tech, Science	2024-01-15, 2024-02-10, 2024-03-05
B002	Python Programming	Williams	Tech	2024-01-20
B003	Database Design	Brown, Davis, Miller	Tech, Academic	2024-02-01, 2024-02-25

Questions:

- Identify ALL attributes that violate 1NF.
- Design the complete set of normalized relations.
- What primary keys would you assign to each new table?

SECOND NORMAL FORM (2NF)

A relation is in 2NF if it is in 1NF and every non-key attribute is fully dependent on the entire primary key (no partial dependencies).

Question 1: Easy

Examine the following table that stores course enrollment information. Identify partial dependencies and decompose to 2NF.

StudentID	CourseID	CourseName	Grade
S001	C101	Database Design	A
S001	C102	Web Development	B
S002	C101	Database Design	A+
S003	C102	Web Development	B+

Questions:
• What is the primary key?
• Identify the partial dependency.
• Decompose to 2NF tables.

Question 2: Medium

The StudentCourse table below tracks which students are enrolled in which courses. The primary key is (StudentID, CourseID). Identify all partial dependencies and normalize to 2NF.

StudentID	CourseID	Semester	InstructorName	Credits	Grade
S001	C101	Fall2024	Dr. Smith	3	A
S001	C102	Fall2024	Dr. Johnson	4	B
S002	C101	Fall2024	Dr. Smith	3	A+
S003	C103	Spring2025	Dr. Brown	3	B+

Tasks:
• List all partial dependencies.
• Create the decomposed 2NF relations.
• Define all primary and foreign keys.

Question 3: Slightly Difficult

A supplier database has the following structure. The primary key is (SupplierID, PartID). Analyze the table for partial dependencies, transitivity issues, and any other anomalies. Decompose it completely to

2NF, showing all intermediate steps and explaining the dependency relationships.

SupplierID	PartID	SupplierName	SupplierCity	PartName	PartCategory	Price	LeadTime
SUP001	P101	TechSupply Inc	New York	Resistor 1K	Electronics	$0.50	7 days
SUP001	P102	TechSupply Inc	New York	Capacitor 10uF	Electronics	$1.20	7 days
SUP002	P101	Components Ltd	Boston	Resistor 1K	Electronics	$0.45	5 days
SUP003	P103	Global Parts	Toronto	LED Red	Electronics	$0.30	10 days

Detailed Tasks:

- Identify the composite primary key and all partial dependencies.
- Show which attributes depend on only part of the key.
- Decompose into 2NF relations with proper key assignments.
- Explain why each decomposition removes a partial dependency.

THIRD NORMAL FORM (3NF)

A relation is in 3NF if it is in 2NF and has no transitive dependencies (non-key attributes must not depend on other non-key attributes).

Question 1: Easy

Examine the Student table below. Identify the transitive dependency and normalize to 3NF.

StudentID	StudentName	DepartmentID	DepartmentName
S001	Alice Johnson	D01	Computer Science
S002	Bob Smith	D02	Business
S003	Carol White	D01	Computer Science
S004	David Brown	D03	Engineering

Questions:
• What is the transitive dependency?
• Decompose to 3NF tables.

Question 2: Medium

The Employee table stores information about employees, their departments, and locations. Identify all transitive dependencies and normalize to 3NF.

EmployeeID	EmployeeName	DeptID	DeptName	DeptLocation	Salary
E001	John Doe	D01	Sales	New York	$60,000
E002	Jane Wilson	D02	IT	Boston	$80,000
E003	Mike Brown	D01	Sales	New York	$62,000
E004	Sarah Davis	D03	HR	Chicago	$55,000

Tasks:
• Trace the transitive dependency path(s).
• Create 3NF relations with all necessary keys.
• Show foreign key relationships.

Question 3: Slightly Difficult

The following OrderDetails table is in 2NF but contains transitive dependencies. The OrderID is the primary key. Identify all transitive dependencies, explain the dependency chains, and decompose the table to 3NF with proper relationship definitions.

	OrderDate	CustomerID	CustomerName	CustomerCity	ProductID	ProductName	CategoryID	CategoryName	Quantity	
	2024-01-15	C001	John Smith	New York	P101	Laptop	CAT01	Electronics	1	
	2024-01-16	C002	Jane Wilson	Boston	P102	Mouse	CAT01	Electronics	3	
	2024-01-17	C001	John Smith	New York	P103	Keyboard	CAT01	Electronics	1	

Detailed Analysis:

- Map out all attribute dependencies (which attributes depend on which others).
- Identify every transitive dependency chain.
- Design the complete 3NF schema with all necessary tables.
- Define primary keys, foreign keys, and explain the relationships.
- Explain why your final design eliminates all transitive dependencies.

Summary Guide

Normal Form	Requirement	Violates If...	Example Anomaly
1NF	Atomic values only	Attributes contain multiple values	Courses field has "Math, Physics, Chemistry"
2NF	In 1NF + No partial dependencies	Non-key attribute depends on part of composite key	CourseName depends only on CourseID, not (StudentID, CourseID)
3NF	In 2NF + No transitive dependencies	Non-key attribute depends on another non-key attribute	DeptName depends on DeptID, which depends on StudentID